Attendance Recovery Computation – Python Problem

Objective: Calculate the current attendance percentage and find the minimum number of consecutive classes a student must attend to reach the required eligibility percentage (75%).

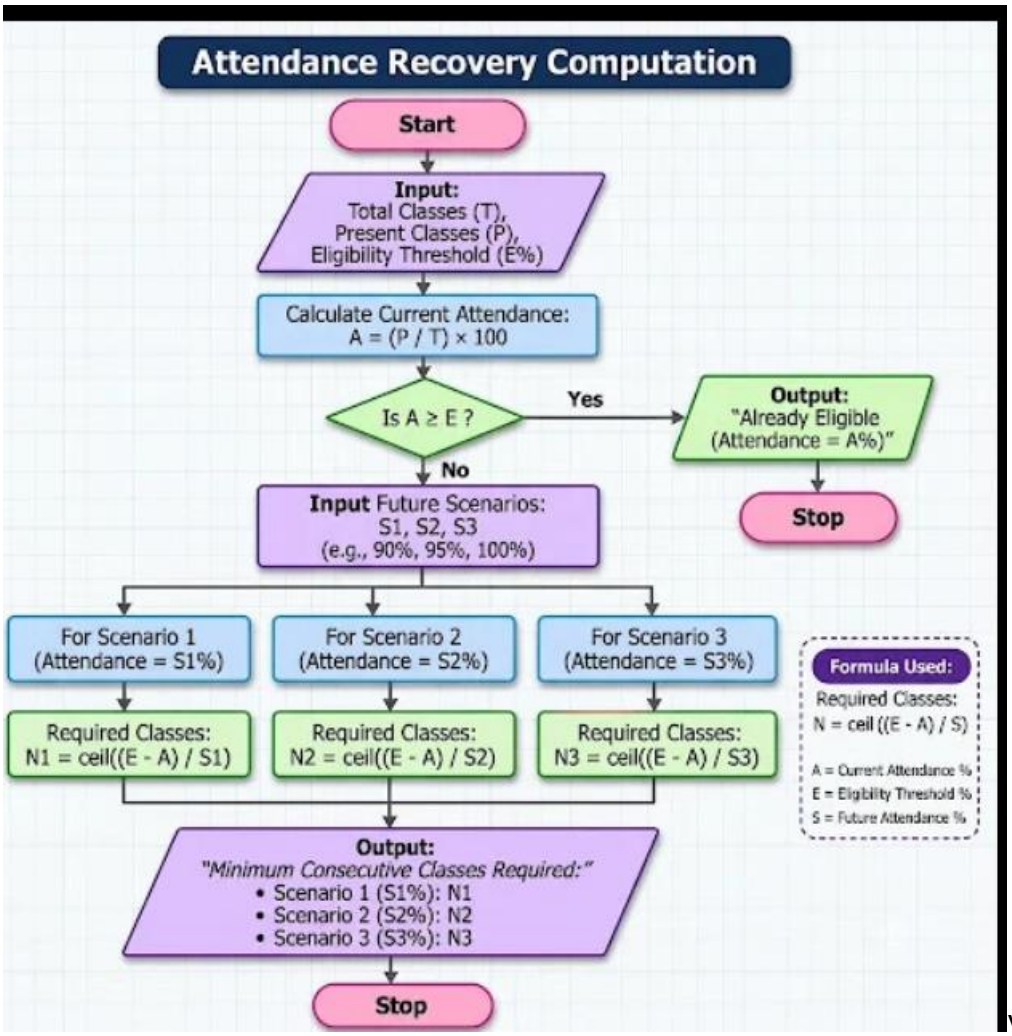
1. Computational Thinking

1. Problem decomposition: Divide the problem into total classes, attended classes, attendance percentage and future classes.
2. Pattern recognition: Attendance increases when a student attends more classes.
3. Abstraction: Focus on total classes, attended classes and eligibility percentage.
4. Algorithm design: Calculate attendance and repeat until the eligibility percentage is reached.

2. Algorithm

1. Start.
2. Read student name.
3. Read total classes conducted.
4. Read classes attended.
5. Calculate current attendance percentage.
6. Display current attendance.
7. Check whether attendance is at least 75%.
8. If yes, display Eligible.

9. Otherwise, calculate the minimum consecuti



ve classes required.

10. Repeat the calculation using a loop.
11. Display the required classes.
12. Stop.

3. Flowchart

4. Variables

Variable	Purpose
name	Student name
total	Total classes
attended	Classes attended

percentage	Current attendance
required	Eligibility percentage
future	Future classes
count	Loop counter

5. Data Types

- str – Student name
- int – Total, attended and future classes
- float – Attendance percentage

6. Operators

- / – Division
- * – Multiplication
- + – Addition
- >= – Greater than or equal to
- += – Increment

7. Conditional Statements

- if – Checks whether attendance is 75% or above.
- else – Executes when attendance is below 75%.

8. Loop

- while loop – Repeats until the attendance reaches 75%.

9. Strings and Pattern Generation

- String: Stores the student's name.
- Pattern generation: Displays a simple attendance progress pattern.

Example:

Attendance Progress:

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10. Lists

Stores student attendance details.

students = ["Ravi", "Kiran", "Arun"]attendance = [70, 80, 65]

11. Tuples and Packing

student = ("Ravi", 60, 40)name, total, attended = student

Here, values are packed into a tuple and unpacked into variables.

12. Pseudocode

START

INPUT name

INPUT total

INPUT attended

required = 75

percentage = (attended / total) * 100

DISPLAY percentage

IF percentage >= required THEN

DISPLAY "Eligible"

ELSE

future = 0

WHILE ((attended + future) /

(total + future)) * 100 < required

future = future + 1

END WHILE

DISPLAY future

END IF

STOP

13. Python Program

name = input("Enter student name: ")total = int(input("Enter total classes: "))attended = int(input("Enter attended classes: "))required = 75if total <= 0 or attended < 0 or attended > total: print("Invalid attendance details")else: percentage = (attended / total) * 100 print("Student:", name) print("Current Attendance:", round(percentage, 2), "%") if percentage >= required: print("Status: Eligible") print("No extra classes required") else: future = 0 while ((attended + future) / (total + future)) * 100 < required: future += 1 print("Status: Not Eligible") print("Minimum classes to attend:", future) print("Attendance Progress:") for i in range(1, 5): print("*" * i)

14. Sample Output

Input:

Enter student name: Ravi

Enter total classes: 100

Enter attended classes: 60

Output:

Student: Ravi

Current Attendance: 60.0 %

Status: Not Eligible

Minimum classes to attend: 60

Attendance Progress:

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15. Different Future Attendance Scenarios

Total classes	Attended	Current %	Extra classes needed
100	60	60%	60
100	70	70%	20
100	75	75%	0
80	50	62.5%	40

Assumption: The student attends every future class.

Conclusion: This program uses computational thinking, algorithms, flowcharts, variables, data types, operators, conditional statements, loops, strings, lists and tuples to calculate attendance eligibility and minimum recovery classes.